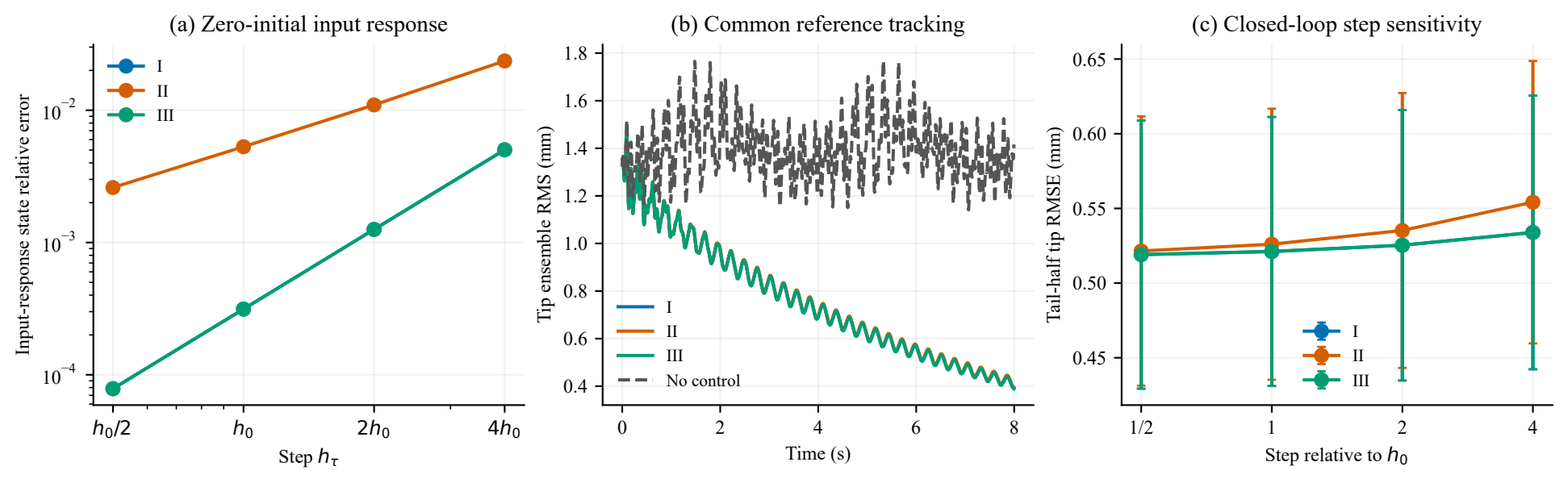

(a) Zero-initial input response
Input-response state relative error
$10^{-2}$
$10^{-3}$
$10^{-4}$
I
II
III
$h_0/2$
$h_0$
$2h_0$
$4h_0$
Step $h_\tau$
(b) Common reference tracking
Tip ensemble RMS (mm)
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
I
II
III
No control
0
2
4
6
8
Time (s)
(c) Closed-loop step sensitivity
Tail-half tip RMSE (mm)
0.65
0.60
0.55
0.50
0.45
I
II
III
1/2
1
2
4
Step relative to $h_0$